

Chapter 9 Test model

- **70 Multiple Choice Questions (MCQs)**
 - **30 True/False Questions**
 - **Difficulty:** 70% Hard, 30% Very Hard
 - **Format:** Clear numbering, blank line (___) between each question
 - **Answer keys** are provided **after all questions**
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Multiple Choice Questions (1–70)

1. In ER-to-relational mapping, what is the primary key of a relation created from a regular entity type?
 - a. Any attribute of the entity
 - b. A surrogate key
 - c. One of the key attributes of the entity
 - d. The foreign key from a related entity
-
2. When mapping a weak entity type, the primary key of the resulting relation consists of:
 - a. Only the partial key of the weak entity
 - b. The primary key of the owner entity only
 - c. The combination of the owner's primary key and the weak entity's partial key
 - d. A newly generated surrogate key
-
3. Which of the following is NOT a valid approach for mapping a binary 1:1 relationship?
 - a. Foreign key in one of the two relations
 - b. Merging both entity types into a single relation
 - c. Creating a separate relation with both primary keys as foreign keys
 - d. Adding a multivalued attribute to one entity
-
4. In mapping a 1:N relationship, where is the foreign key placed?
 - a. In the relation corresponding to the '1' side
 - b. In the relation corresponding to the 'N' side
 - c. In a new relationship relation
 - d. In both relations

-
5. For an M:N relationship, the standard mapping technique requires:
- a. Adding a foreign key to one entity relation
 - b. Merging both entities
 - c. Creating a new relation with foreign keys from both entities
 - d. Using a type attribute
-

6. The primary key of a relationship relation for an M:N binary relationship is:
- a. The primary key of the first entity
 - b. The primary key of the second entity
 - c. A surrogate key
 - d. The combination of both foreign keys
-

7. How is a multivalued attribute mapped in the relational model?
- a. As a simple attribute in the same relation
 - b. As a separate relation with the entity's primary key as a foreign key
 - c. By converting it into a derived attribute
 - d. By ignoring it during mapping
-

8. The primary key of the relation created for a multivalued attribute includes:
- a. Only the multivalued attribute
 - b. Only the foreign key from the owning entity
 - c. Both the multivalued attribute and the foreign key
 - d. A surrogate key
-

9. For an n-ary relationship ($n > 2$), the mapping produces:
- a. A single merged relation of all participating entities
 - b. A new relation with foreign keys from all participating entity relations
 - c. A foreign key added to the largest entity relation
 - d. No relation—n-ary relationships cannot be mapped
-

10. Which mapping option for specialization requires that every entity in the superclass belongs to at least one subclass?
- a. Option 8A

- b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

11. In Option 8A for mapping specialization, the subclass relations contain:
- a. Only subclass-specific attributes
 - b. All superclass attributes plus subclass attributes
 - c. Only the primary key of the superclass
 - d. The primary key and subclass-specific attributes
-

12. Option 8C uses what kind of attribute to indicate subclass membership?
- a. Surrogate key
 - b. Boolean flag
 - c. Type (discriminating) attribute
 - d. Partial key
-

13. Option 8D is most suitable for which type of specialization?
- a. Total and disjoint
 - b. Partial and disjoint
 - c. Overlapping (non-disjoint)
 - d. Attribute-defined only
-

14. In mapping a shared subclass (multiple inheritance), what must be true about the superclasses?
- a. They must have the same primary key
 - b. They must be weak entities
 - c. They must participate in a 1:1 relationship
 - d. They must have disjoint attributes
-

15. What is used to map a category (union type) when the superclasses have different keys?
- a. Composite key
 - b. Partial key
 - c. Surrogate key
 - d. Type attribute

16. In the COMPANY database example, the DEPENDENT relation's primary key is:

- a. SSN
- b. DEPENDENT_NAME
- c. {ESSN, DEPENDENT_NAME}
- d. {SSN, RELATIONSHIP}

17. In Step 4 of the ER-to-relational mapping, attributes of the 1:N relationship itself are placed in:

- a. The '1' side relation
- b. The 'N' side relation
- c. A new relationship relation
- d. Both relations

18. Which of the following relationships in the COMPANY schema is M:N?

- a. MANAGES
- b. WORKS_FOR
- c. SUPERVISION
- d. WORKS_ON

19. The relation DEPT_LOCATIONS is created due to which ER construct?

- a. Weak entity
- b. Multivalued attribute
- c. M:N relationship
- d. Composite attribute

20. In Option 8B, the superclass relation is:

- a. Created with all attributes
- b. Not created at all
- c. Created only with the key
- d. Merged into a view

21. What is a major limitation of the relational model regarding ER constraints?

- a. It cannot represent entities

- b. It cannot enforce maximum cardinality ratios like 1:10
 - c. It cannot store string data
 - d. It cannot handle foreign keys
-

22. Which mapping strategy minimizes null values in specialization hierarchies when subclasses are disjoint and total?
- a. Option 8A
 - b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

23. In Figure 9.7 (mapping categories), the OWNER relation uses:
- a. Composite key from BANK and PERSON
 - b. Surrogate key OwnerId
 - c. Boolean flags
 - d. Type attribute
-

24. The foreign key in the EMPLOYEE relation representing WORKS_FOR is named:
- a. DNUMBER
 - b. DEPT_NO
 - c. DNO
 - d. MANAGER_SSN
-

25. Which of the following is a goal of the ER-to-relational mapping algorithm?
- a. Maximize redundancy
 - b. Preserve all information and minimize nulls
 - c. Eliminate all foreign keys
 - d. Convert all attributes to multivalued
-

26. In Step 3 (1:1 mapping), why is it better to place the foreign key in the entity with total participation?
- a. To avoid nulls
 - b. To reduce relations
 - c. To simplify queries
 - d. To enforce disjointness

27. For a weak entity with no partial key, the primary key of its relation is:

- a. A surrogate key
 - b. The owner's primary key alone
 - c. Not defined
 - d. A composite of all attributes
-

28. The relationship relation for SUPPLY (ternary) has a primary key consisting of:

- a. One foreign key
 - b. Two foreign keys
 - c. Three foreign keys
 - d. A surrogate key
-

29. Which of the following is NOT preserved in relational mapping?

- a. Entity attributes
 - b. Participation constraints (total/partial)
 - c. Key constraints
 - d. Exhaustive classification in specialization
-

30. In Option 8C, if a specialization is overlapping, what issue arises?

- a. The type attribute can only hold one value
 - b. Nulls increase
 - c. Foreign keys conflict
 - d. Primary key becomes composite
-

31. What distinguishes a category from a shared subclass?

- a. Categories have same-key superclasses
 - b. Shared subclasses have different-key superclasses
 - c. Categories have different-key superclasses
 - d. Categories cannot have attributes
-

32. In mapping EER models, which option allows an entity to belong to multiple subclasses simultaneously?

- a. Option 8A

- b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

33. The MANAGES relationship in COMPANY is mapped using which approach?
- a. Merged relation
 - b. Foreign key in DEPARTMENT
 - c. Foreign key in EMPLOYEE
 - d. Separate relationship relation
-

34. If a multivalued composite attribute exists, how is it mapped?
- a. Ignored
 - b. As a single attribute
 - c. As simple components in a new relation
 - d. As a derived attribute
-

35. Which step handles attributes like {Phone1, Phone2, Phone3} if modeled as multivalued?
- a. Step 1
 - b. Step 2
 - c. Step 6
 - d. Step 5
-

36. In Step 7, for an n-ary relationship, simple attributes of the relationship are:
- a. Dropped
 - b. Added to one entity relation
 - c. Included in the new relationship relation
 - d. Converted to foreign keys
-

37. What is the main drawback of Option 8C for overlapping specializations?
- a. Wasted storage
 - b. Inability to represent multiple subclass membership
 - c. Complex joins
 - d. Loss of key constraints

38. In the relational schema, how is total participation in a 1:N relationship enforced?

- a. By NOT NULL on foreign key
- b. By unique constraint
- c. It cannot be fully enforced
- d. By check constraint on type

39. Which of the following is mapped to a relation with a composite primary key by default?

- a. Regular entity
- b. Weak entity
- c. 1:1 relationship with total participation
- d. Multivalued attribute

40. The attribute HOURS in WORKS_ON comes from:

- a. EMPLOYEE
- b. PROJECT
- c. The WORKS_ON relationship type
- d. A derived calculation

41. In Option 8D, how many Boolean flags are needed for 4 subclasses?

- a. 1
- b. 2
- c. 4
- d. 0

42. A shared subclass like STUDENT_ASSISTANT inherits:

- a. Different keys from each superclass
- b. The same key from all superclasses
- c. A new surrogate key
- d. No key

43. Which mapping option is most storage-efficient for a total, disjoint specialization?

- a. Option 8A
- b. Option 8B

- c. Option 8C
 - d. Option 8D
-

44. In Figure 9.5(c), the type attribute is named:

- a. JobType
 - b. EmployeeType
 - c. SubclassID
 - d. Category
-

45. The foreign key in DEPENDENT referencing EMPLOYEE is named:

- a. SSN
 - b. ESSN
 - c. EMP_ID
 - d. DEP_SSN
-

46. What is the primary reason to avoid the "relationship relation" approach for 1:N relationships?

- a. It increases nulls
 - b. It's redundant and inefficient
 - c. It violates normalization
 - d. It cannot store attributes
-

47. In Step 1, composite keys are handled by:

- a. Creating a surrogate key
 - b. Using all simple components as part of the primary key
 - c. Dropping non-key components
 - d. Converting to multivalued
-

48. Which construct requires a new relation even if it has no attributes?

- a. Weak entity
 - b. M:N relationship
 - c. 1:1 relationship
 - d. Multivalued attribute
-

49. The DEPT_LOCATIONS relation has which primary key?

- a. DLOCATION
 - b. DNUMBER
 - c. {DNUMBER, DLOCATION}
 - d. LOC_ID
-

50. In EER mapping, if a superclass has 1000 entities but only 100 belong to any subclass, which option minimizes nulls?

- a. Option 8A
 - b. Option 8B
 - c. Option 8C
 - d. Option 8D
-

51. Which of the following cannot be directly enforced in the relational model after ER mapping?

- a. Key constraints
 - b. Referential integrity
 - c. Disjointness constraint in specialization
 - d. Domain constraints
-

52. In Option 8A, how is referential integrity between superclass and subclass maintained?

- a. Foreign key from subclass to superclass
 - b. Same primary key value
 - c. Trigger
 - d. View
-

53. The main advantage of Option 8D over 8C is:

- a. Less storage
 - b. Supports overlapping subclasses
 - c. Simpler queries
 - d. No nulls
-

54. In the SUPPLY ternary relationship, if one of the participating entities has a composite key, how is it handled in the relationship relation?

- a. Only one part is used

- b. All simple components are included as foreign keys
 - c. Replaced by surrogate
 - d. Ignored
-

55. Which step maps a relationship like "Teaches" between Professor, Course, and Semester (3-way)?
- a. Step 4
 - b. Step 5
 - c. Step 7
 - d. Step 3
-

56. If an entity has a derived attribute, how is it treated in relational mapping?
- a. Stored as a regular attribute
 - b. Not mapped (computed at query time)
 - c. Mapped as multivalued
 - d. Requires a new relation
-

57. In mapping a 1:1 relationship with partial participation on both sides, which approach is safest?
- a. Merge into one relation
 - b. Foreign key in either relation
 - c. Separate relationship relation
 - d. Ignore the relationship
-

58. What is the biggest risk of using Option 8B for a partial specialization?
- a. Data duplication
 - b. Loss of superclass entities not in any subclass
 - c. Null values
 - d. Key conflicts
-

59. In Figure 9.7, the STUDENT_ASSISTANT relation includes keys from:
- a. STUDENT and EMPLOYEE
 - b. GRADUATE and EMPLOYEE
 - c. PERSON and DEPARTMENT
 - d. Only STUDENT

60. Categories are mapped similarly to:

- a. Weak entities
 - b. Strong entities with surrogate keys
 - c. M:N relationships
 - d. Multivalued attributes
-

61. Which of the following is true about the foreign key in a weak entity relation?

- a. It can be null
 - b. It must match the owner's primary key and cannot be null
 - c. It is optional
 - d. It is always composite
-

62. In Step 6, if a multivalued attribute is also composite, the new relation includes:

- a. Only the composite name
 - b. The simple components of the composite attribute
 - c. A surrogate key only
 - d. Nothing—it's not mappable
-

63. The mapping algorithm prioritizes:

- a. Minimizing relations
 - b. Preserving information and minimizing nulls
 - c. Eliminating foreign keys
 - d. Using surrogate keys everywhere
-

64. In Option 8C, what happens if an entity belongs to two subclasses in an overlapping hierarchy?

- a. Two tuples are created
 - b. Only one subclass is recorded
 - c. An error occurs
 - d. Boolean flags are used instead
-

65. Which ER construct is the only one that always produces a relation with a composite primary key made of foreign keys?

- a. Weak entity
 - b. 1:1 relationship
 - c. M:N relationship
 - d. Multivalued attribute
-

66. In the COMPANY schema, the SUPERVISION relationship is:

- a. M:N
 - b. 1:N
 - c. 1:1
 - d. Ternary
-

67. The attribute DNO in EMPLOYEE is a:

- a. Primary key
 - b. Partial key
 - c. Foreign key
 - d. Derived attribute
-

68. For a weak entity with a composite partial key, the primary key of its relation is:

- a. The owner's key only
 - b. The partial key only
 - c. Owner's key + all simple parts of partial key
 - d. Surrogate key
-

69. In Step 3, the "merged relation" option is best when:

- a. Participation is partial on both sides
 - b. Both participations are total
 - c. One side is weak
 - d. The relationship has attributes
-

70. Which of the following is a valid reason to use a surrogate key in category mapping?

- a. Superclasses have different primary keys
- b. To reduce storage
- c. To avoid composite keys
- d. All of the above

True/False Questions (71–100)

71. In ER-to-relational mapping, every relationship type must result in a new relation.

- A. True
 - B. False
-

72. A weak entity's relation always includes the owner's primary key as a foreign key.

- A. True
 - B. False
-

73. Option 8B can be used for partial specializations.

- A. True
 - B. False
-

74. Multivalued attributes are stored in the same relation as the entity in the relational model.

- A. True
 - B. False
-

75. The primary key of an M:N relationship relation is always a surrogate key.

- A. True
 - B. False
-

76. In Option 8C, the type attribute can represent multiple subclass memberships.

- A. True
 - B. False
-

77. Categories require a surrogate key only when superclasses have different keys.

- A. True
 - B. False
-

78. The relational model can fully enforce total participation constraints from ER diagrams.

- A. True
 - B. False
-

79. In Step 4, the foreign key is placed on the '1' side of a 1:N relationship.

- A. True
 - B. False
-

80. A shared subclass must have the same key attribute across all its superclasses.

- A. True
 - B. False
-

81. Option 8D uses a single type attribute to indicate subclass membership.

- A. True
 - B. False
-

82. The `DEPENDENT_NAME` in the `DEPENDENT` relation is a partial key.

- A. True
 - B. False
-

83. All attributes of a superclass are duplicated in every subclass relation in Option 8A.

- A. True
 - B. False
-

84. N-ary relationships are mapped by creating a new relation with foreign keys from all participating entities.

- A. True
 - B. False
-

85. The `MANAGES` relationship in `COMPANY` is 1:N.

- A. True

B. False

86. In Option 8B, no relation is created for the superclass.

- A. True
 - B. False
-

87. A derived attribute is always stored in the relational schema.

- A. True
 - B. False
-

88. The goal of minimizing null values is always achievable in ER-to-relational mapping.

- A. True
 - B. False
-

89. Composite attributes are stored as a single field in the relational model.

- A. True
 - B. False
-

90. In Step 6, the primary key of the new relation for a multivalued attribute includes the entity's primary key and the attribute value.

- A. True
 - B. False
-

91. Overlapping specializations can be correctly represented using Option 8C.

- A. True
 - B. False
-

92. The WORKS_ON relation has a primary key of {ESSN, PNO}.

- A. True
 - B. False
-

93. In Step 3, the foreign key approach for 1:1 relationships can lead to nulls if participation is partial.

- A. True
 - B. False
-

94. A category is a type of specialization.

- A. True
 - B. False
-

95. The relational model can preserve exhaustive classification constraints from EER diagrams.

- A. True
 - B. False
-

96. In Option 8A, subclass relations have the same primary key as the superclass.

- A. True
 - B. False
-

97. The SUPPLY relationship relation's primary key includes SNAME, PARTNO, and PROJNAME.

- A. True
 - B. False
-

98. Weak entities can exist without an owner entity in the relational model.

- A. True
 - B. False
-

99. All relationship attributes in 1:N relationships are placed in the 'N' side relation.

- A. True
 - B. False
-

100. Step 7 applies only to ternary relationships, not higher n-ary ones.

- A. True
 - B. False
-

Answer Key

Multiple Choice (1–70)

- 1. c
- 2. c
- 3. d
- 4. b
- 5. c
- 6. d
- 7. b
- 8. c
- 9. b
- 10. b
- 11. d
- 12. c
- 13. c
- 14. a
- 15. c
- 16. c
- 17. b
- 18. d
- 19. b
- 20. b
- 21. b
- 22. b
- 23. b
- 24. c
- 25. b
- 26. a
- 27. b
- 28. c
- 29. d
- 30. a
- 31. c

32. d

33. b

34. c

35. c

36. c

37. b

38. c

39. b

40. c

41. c

42. b

43. b

44. a

45. b

46. b

47. b

48. b

49. c

50. a

51. c

52. b

53. b

54. b

55. c

56. b

57. c

58. b

59. a

60. b

61. b

62. b

63. b

64. b

65. c

66. b

67. c

68. c

69. b

70. a

True/False (71–100)

- 71. B (False) — only M:N and n-ary always create new relations
 - 72. A (True)
 - 73. B (False) — only for total
 - 74. B (False) — requires separate relation
 - 75. B (False) — it's the combo of foreign keys
 - 76. B (False) — only one value allowed
 - 77. A (True)
 - 78. B (False) — only partially via NOT NULL
 - 79. B (False) — on the 'N' side
 - 80. A (True)
 - 81. B (False) — that's Option 8C; 8D uses multiple Booleans
 - 82. A (True)
 - 83. B (False) — only the key is in subclass relations in 8A
 - 84. A (True)
 - 85. B (False) — it's 1:1
 - 86. A (True)
 - 87. B (False) — usually not stored
 - 88. B (False) — not always possible
 - 89. B (False) — broken into simple attributes
 - 90. A (True)
 - 91. B (False) — cannot represent multiple membership
 - 92. A (True)
 - 93. A (True)
 - 94. B (False) — category is union, not specialization
 - 95. B (False) — cannot enforce exhaustive classification
 - 96. A (True)
 - 97. A (True)
 - 98. B (False) — must reference owner
 - 99. A (True)
 - 100. B (False) — applies to any $n > 2$
-